Reflective Portfolio Report

Ethical Analysis of the Epic Sepsis Model

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Chapter 1: **Introduction**

Sepsis is a life-threatening condition caused by a dysregulated host response to infection and is associated with high mortality and substantial healthcare costs. In hospitalized patients, sepsis occurs in approximately 2–6% of admissions, with reported mortality rates ranging from 15% to 40% depending on severity and the time of intervention.

Clinical evidence consistently shows that earlier identification and treatment, particularly timely administration of antibiotics, significantly reduce mortality. As a result, healthcare systems have increasingly adopted AI-enabled Clinical Decision Support Systems (CDSS) to assist clinicians in early detection and risk stratification of sepsis.

This project focuses on a real-world, deployed AI-enabled CDSS: the **Epic Sepsis Model (ESM)** Inpatient Predictive Analytic Tool, integrated into the Epic electronic health record (EHR) system. Epic Systems is the largest EHR provider in the United States, serving approximately 54% of patients nationwide, making the ESM one of the most widely deployed AI-based decision support tools in clinical practice.

The ESM is designed to continuously monitor hospitalized patients and generate a numerical risk score indicating the likelihood of sepsis, enabling earlier clinical intervention.

Unlike experimental or research-only models, the ESM operates in live hospital environments and directly influences clinical workflows. Its outputs trigger alerts to nurses and physicians when a predefined risk threshold is exceeded, thereby shaping diagnostic attention, treatment prioritisation, and escalation decisions. The ESM therefore represents a *safety-critical* AI system whose ethical implications extend beyond model accuracy to issues of trust, accountability, over-reliance, and human–AI collaboration.

Key Features of the System

The Epic Sepsis Model is a data-driven, sub-symbolic AI system implemented within a commercial electronic health record platform. Published validation studies provide sufficient evidence to characterise its technical structure and behaviour.

AI Technology and Model Characteristics

Property	Detail
Model type	Data-driven predictive model (sub-symbolic)
Input features	≈80 clinical and demographic variables
	(vital signs, lab results, comorbidities, demographics)
Training data	≈500,000 patient encounters
Output	Continuously updated numerical sepsis risk score
Deployment	Fully integrated into Epic EHR; operates in real time

Table 1.1: Summary of Epic Sepsis Model technical characteristics.

The ESM does not rely on explicit clinical rules, symbolic reasoning, or hand-crafted knowledge representations. Patient data are combined and weighted according to learned statistical relationships to produce a risk score. While the system can surface which variables contributed most to an elevated score, its internal decision logic is not fully transparent to end users, placing it firmly in the category of *sub-symbolic, black-box* clinical decision support.

Operational Threshold and Performance

In the validation study conducted at a 746-bed academic medical centre, an alert threshold of **ESM** $\geq$ **5** was selected using receiver operating characteristic (ROC) curve analysis. Table 1.2 summarises the performance at this threshold.

Metric	**Value**
Area Under ROC Curve (AUC)	0.834
Sensitivity	86.0%
Specificity	80.8%
Positive Predictive Value (PPV)	33.8%
Negative Predictive Value (NPV)	98.1%

Table 1.2: ESM performance at alert threshold ≥ 5.

These results indicate that the ESM is optimised as a *screening* tool, prioritising sensitivity and early detection rather than definitive diagnosis.

1.1 Functional Goals of the System

The primary functional goal of the Epic Sepsis Model is the early identification of hospitalized patients at risk of developing sepsis, enabling clinicians to initiate timely evaluation and treatment. This goal is operationalised through continuous risk scoring and automated alerting within the EHR.

1. Detect elevated sepsis risk earlier than routine clinical recognition. By continuously analysing patient data, the ESM aims to identify deterioration before overt clinical signs prompt clinician action.

2. Prompt timely clinical intervention. When the ESM score exceeds the alert threshold, clinicians are notified and expected to assess the patient, consider sepsis protocols, and initiate appropriate management.

3. Reduce time to antibiotic administration. In the post-implementation phase of the validation study, the median time from alert to antibiotic administration decreased from 150 minutes to 90 minutes, and the proportion of patients receiving antibiotics within three hours increased from 55.6% to 69.8%.

4. Improve patient outcomes, particularly mortality. Among patients with ESM scores ≥ 5 who had not already received antibiotics, implementation was associated with a **44% reduction** in the odds of in-hospital sepsis-related mortality (odds ratio 0.56; 95% CI 0.39–0.80).

5. Integrate seamlessly into existing clinical workflows. The system is embedded within the Epic EHR and requires no separate interface, enabling widespread deployment with minimal disruption.

Ethical Relevance of the Functional Design

Although the ESM demonstrably improves early intervention and is associated with reduced mortality, its design raises ethical challenges beyond functional performance. The system produces risk scores that significantly influence clinician behaviour, yet its internal reasoning is largely opaque. Clinicians remain legally and morally responsible for all decisions, while the AI system shapes attention, urgency, and treatment timing. This creates potential risks of *automation bias*, *over-reliance*, and *accountability gaps* in high-stakes clinical contexts. These considerations motivate the focus of this project: not on building a new predictive model, but on explicitly identifying ethical goals, formalising them as constraints or checks, and demonstrating through targeted scripts and validation cases when such goals may or may not be satisfied.

Chapter 2: **Stakeholder Identification**

The Epic Sepsis Model (ESM) operates within a complex hospital environment involving clinical personnel, patients, healthcare institutions, and technology providers. As a deployed AI-enabled Clinical Decision Support System (CDSS), its ethical impact arises from interactions between these stakeholders rather than from the predictive model alone. This chapter identifies the primary and secondary stakeholders affected by the system and maps them across social, technical, economic, environmental, political and legal dimensions.

2.1 Stakeholders of the Epic Sepsis Model

Primary Stakeholders

Primary stakeholders are those who directly interact with the system or are directly affected by its outputs.

1. Clinicians (Physicians and Nurses)

Category: Social, Technical, Legal
Clinicians are the primary users of the ESM. The system generates alerts when a patient's sepsis risk score exceeds a predefined threshold ($\mathbf{ESM} \geq \mathbf{5}$), prompting clinical evaluation and intervention. Although clinicians retain final decision-making authority, the alerts influence attention, prioritisation, and timing of treatment. Clinicians are also legally accountable for patient outcomes, even when decisions are informed by AI-generated risk scores.

2. Patients

Category: Social, Ethical, Legal
Hospitalised patients are the subjects of the predictive model. They do not directly interact with the system but are affected by its outputs through changes in clinical behaviour, including earlier antibiotic administration and ICU escalation. Patients bear the risks of false positives (unnecessary treatment, antibiotic overuse) and false negatives (missed or delayed diagnosis).

3. Hospital and Healthcare Providers

Category: Economic, Technical, Legal
Hospitals deploy the ESM as part of the Epic EHR infrastructure. They benefit from potential reductions in mortality, length of stay, and compliance with sepsis quality measures. At the same time, hospitals assume institutional liability and must manage alert fatigue, workflow disruption, and training requirements.

Secondary Stakeholders

Secondary stakeholders influence the system indirectly or are affected at a broader systemic level.

4. Epic Systems Corporation

Category: Technical, Economic, Legal
Epic develops and maintains the proprietary ESM algorithm. Its design choices including feature selection, model updates and alert presentation shape how clinicians interpret and act on risk scores.

5. Health Regulators and Accreditation Bodies

Category: Political, Legal

Regulatory bodies influence adoption of sepsis detection systems through quality standards, reporting requirements, and patient safety regulations. Although the ESM is not publicly regulated as a medical device, its use affects compliance with national sepsis guidelines.

6. Society and Public Health Systems

Category: Social, Economic
At a population level, widespread deployment of sepsis detection tools influences healthcare costs, antibiotic stewardship, and public trust in AI-assisted medicine.

7. Environmental and Infrastructure Considerations

Category: Environmental, Technical
Large scale deployment across hospital networks contributes to cumulative energy consumption and infrastructure expansion. While individual risk score computations have negligible marginal environmental impact, responsible digital health governance must consider long-term sustainability of computational infrastructure.

2.2 Ethical Goals of the System

The Epic Sepsis Model does not publish an explicit ethical framework or set of ethical principles. Public documentation and validation studies focus primarily on clinical effectiveness, workflow integration, and patient outcomes.

There are no explicit public statements addressing transparency, accountability, or ethical safeguards beyond standard clinical governance practices.

This absence is ethically relevant, as it requires ethical safeguards to be inferred from system behaviour and deployment structure rather than from explicitly articulated normative commitments.

2.3 Ethical Considerations and Schools of Thought

For this project, three ethical frameworks are used to analyse the system:

- **Utilitarianism**
- **Deontological Ethics**
- **Virtue Ethics**

These were selected because they capture outcome-based, duty-based, and professional character-based ethical concerns relevant to clinical AI systems.

1. Utilitarian Ethics (Outcome-Oriented)

Core Principle: Actions are ethically justified if they maximise overall benefit and minimise harm across all affected parties (Bentham, Mill).

Relevant Variables and System Functionality. Sepsis mortality rates across the patient population; time to antibiotic administration per alert; false positive rate (FPR) and its burden on clinical workflow; hospital-wide resource utilisation.

Ethical Goal U1: Minimise Preventable Sepsis Mortality.

Actionable Principle (Utilitarian: Greatest Aggregate Benefit): The ESM alert system is ethically justified only if the reduction in sepsis-related mortality across the treated population outweighs the aggregate harm caused by false positive alerts, including unnecessary antibiotic administration, increased costs, and alert fatigue-induced inattention. Vasey et al. [1] (DECIDE-AI) demonstrate that ethical risks in AI-based CDSS frequently arise from deployment failures rather than model errors, meaning the utilitarian calculation must be evaluated continuously in real-world conditions, not only at the point of validation.

Measurable Condition: The system satisfies this principle if the odds ratio of sepsis-related mortality in the alert-active period remains below 1.0 at a statistically significant level ($p < 0.05$). The published validation study [2] reports OR = 0.56 (95% CI 0.39–0.80), satisfying this condition at the studied institution.

Violation Trigger: If post-deployment monitoring reveals that the false positive burden has increased clinician override rates to the point where true positive alerts are being systematically ignored, the net utility calculation is violated. Specifically, if the proportion of true positive alerts acted upon falls below the proportion in the pre-implementation period, the system no longer produces net benefit and the utilitarian justification for its deployment fails.

Ethical Goal U2: Control False Positive Burden.

Actionable Principle (Utilitarian : Harm Minimisation): The false positive rate must be controlled such that the cumulative harm of unnecessary interventions does not outweigh the benefit of early detection. A system with sensitivity optimised at 86.0% and PPV of 33.8% accepts that approximately two in three alerts are false positives [2]. This trade-off is ethically acceptable only if alert fatigue does not reduce the clinical response rate to true positive alerts. Shwedeh and Alzoubi [3] demonstrate empirically that over-reliance and alert fatigue emerge specifically in high-pressure deployment environments where governance and oversight are insufficient, reinforcing that this condition must be actively monitored rather than assumed.

Measurable Condition: Alert fatigue is operationally defined as a clinician override rate exceeding 70% across consecutive alert sessions. If this threshold is crossed, the PPV of acted-upon alerts falls below the level required for net population benefit.

Violation Trigger: Override rate > 70% across a rolling window of alerts, or a statistically significant increase in time-to-antibiotic following the introduction of the alert system.

2. Deontological Ethics (Duty and Responsibility)

Core Principle: Ethical action is determined by adherence to duties and rules, independent of outcomes. Agents must be treated as rational ends in themselves, not merely as means (Kant, *Groundwork of the Metaphysics of Morals*).

Relevant Variables and System Functionality. Availability of model reasoning to clinicians; clinician legal and moral accountability for alert-informed decisions; governance requirement for alert acknowledgement.

Ethical Goal D1: Preserve Clinician Epistemic Access (Duty of Non-Deception).

Actionable Principle (Kantian : Respect for Rational Agency): A clinician who is held legally and morally responsible for a clinical decision must have meaningful epistemic access to the basis of the information that prompted that decision. Denying access to model reasoning while retaining accountability violates the Kantian duty to treat rational agents as ends in themselves, the clinician is reduced to a conduit for automated output rather than a deliberating professional. Amann et al. [4] establish that lack of explainability causes systems nominally classified as decision support to drift toward decision determining, undermining meaningful human oversight. Xu et al. [5] further argue that interpretability is a fundamental ethical requirement for AI-based CDSS, not an optional feature, and that post-hoc explanation methods alone are insufficient to guarantee accountability. Čartolovni et al. [6] identify the resulting accountability gap where formal responsibility remains with the clinician while practical epistemic control has shifted to the AI as the dominant ethical concern in the AI-based medical decision support literature.

Measurable Condition: The system satisfies this principle if, for every alert fired, the clinician can access at minimum the top contributing feature variables and their directional influence on the score. The current ESM does not satisfy this condition, feature-level reasoning is not exposed to bedside clinicians.

Violation Trigger: Any alert fired without accompanying feature-level explanation constitutes a violation of this principle. In the current deployment, this condition is violated for 100% of alerts.

Ethical Goal D2: Enforce Governance Accountability (Duty of Institutional Responsibility).

Actionable Principle (Deontological : Rule-Based Governance): Every alert fired by the system must be formally acknowledged by a responsible clinician. An unacknowledged alert represents a failure of the institutional duty of care, the system has identified a patient at risk and no accountable agent has accepted responsibility for the assessment.

Measurable Condition: Governance is satisfied if and only if for all alerts A fired by the system, there exists a corresponding clinician acknowledgement C within a defined response window T. Formally: $Alert(A) \rightarrow Acknowledged(C, T)$.

Violation Trigger: Any alert that passes without documented clinician acknowledgement within the response window violates this rule. This is operationalised as the `governance_check` constraint in the implementation and verified formally using Z3 SMT.

3. Virtue Ethics (Professional Character and Practice)

Core Principle: Ethical systems should cultivate and support good professional character such as practical wisdom (*phronesis*), attentiveness, and calibrated judgement rather than replace them (Aristotle, *Nicomachean Ethics*).

Relevant Variables and System Functionality. Clinician override and compliance patterns over time; alert frequency and its effect on attentiveness; long-term effect of AI-assisted workflows on clinical skill.

Ethical Goal V1: Prevent Automation Bias and Preserve Clinical Prudence.

Actionable Principle (Virtue Ethics : Phronesis): A clinical AI system must be designed and deployed in a way that supports rather than displaces the exercise of practical wisdom. Automation bias: the tendency to follow AI output without independent assessment directly undermines *phronesis*. A system that produces this effect at scale is ethically deficient regardless of its population-level mortality benefit. Panigutti et al. [7] provide empirical evidence that AI explanations systematically increase human reliance on AI advice regardless of whether that reliance improves decision accuracy, demonstrating that the problem of automation bias cannot be resolved by adding explanations alone. Čartolovni et al. [6] identify automation bias and deskilling as the most clinician-specific ethical risks in AI-based medical decision support, directly implicating virtue ethics as the relevant framework.

Measurable Condition: Automation bias is operationally present if the clinician compliance rate with alerts defined as the proportion of alerts followed without documented independent clinical assessment exceeds 90% across a rolling observation window.

Violation Trigger: Compliance rate $>$ 90% without documented independent assessment, operationalised as the `detect_over_reliance` constraint in the implementation (threshold: `compliance_rate > 0.9`).

Ethical Goal V2: Prevent Clinical Deskilling.

Actionable Principle (Virtue Ethics : Maintenance of Professional Competence): Repeated reliance on AI-generated alerts without active clinical reasoning constitutes a long-term threat to the professional competence required for safe independent practice. Clinicians who have not practiced independent sepsis assessment over an extended period may lose the diagnostic calibration that makes meaningful human oversight possible. Čartolovni et al. [6] explicitly identify deskilling and loss of epistemic authority as physician-specific ethical risks in AI-based decision support, distinct from automation bias a clinician may not be biased in any given moment but may have cumulatively lost the skill required for genuine independent assessment.

Measurable Condition: Deskilling risk is flagged if clinician independent override decisions, cases where a clinician actively disagrees with and documents a reasoned rejection of an alert fall below a defined minimum frequency per clinician per month.

Violation Trigger: Independent override rate $<$ minimum threshold per clinician per defined period, indicating habitual passive compliance rather than active clinical engagement.

Ethical Framework–Stakeholder Alignment

The three ethical schools correspond to different stakeholder perspectives. Utilitarian analysis primarily concerns patient outcomes and hospital-level mortality reduction. Deontological considerations are most salient for clinicians, who retain formal decision authority and legal responsibility for AI-informed actions. Virtue ethics focuses on the professional character of clinicians interacting repeatedly with alert systems, emphasising calibration, attentiveness, and avoidance of automation bias. Together, these frameworks provide a pluralistic ethical lens appropriate for socio-technical clinical systems.

2.4 Values

Stakeholder Values and Alignment

Stakeholder	Primary Values	Alignment with Ethical Goals	Conflict
Clinicians	Professional autonomy, accountability, patient safety	D1, D2, V1, V2	Conflicts with Epic's proprietary opacity (D1)
Patients	Timely treatment, equitable care, safety	U1, U2, D1	No direct conflict benefit from all goals
Hospitals	Efficiency, liability reduction, compliance	U1, U2, D2	Cost of explainability features conflicts with U2
Epic Systems	Proprietary protection, commercial viability	None stated publicly	Directly conflicts with D1 (transparency)
Regulators	Safety, standardisation, accountability	D2, U1	May conflict with speed of deployment
Society	Equitable access, antibiotic stewardship	U2, V1	Alert fatigue burden conflicts with U2

Table 2.1: Stakeholder values, ethical goal alignment, and conflicts.

Values Requiring Negotiation

The most significant value conflict is between Epic's commercial interest in maintaining proprietary opacity and the deontological requirement for clinician epistemic access (D1). These cannot be simultaneously satisfied in the current deployment. Resolving this conflict requires either mandatory explainability regulation or contractual transparency obligations in hospital procurement.

A secondary conflict exists between the utilitarian justification for high sensitivity (U1) and the harm minimisation requirement (U2). Maximising sensitivity necessarily increases false positive burden. This trade-off cannot be resolved by system design alone, it requires institutional threshold governance involving clinicians, patients and ethics review.

Prioritisation

Patient safety values take precedence, as patients are the population most directly harmed by system failures and least able to advocate within the deployment structure. Clinician professional values are second, as they are the responsible agents whose judgement the system most directly affects. Institutional and commercial values, while legitimate, must not override the first two categories.

Functional-to-Ethical Goal Balance

The primary functional purpose of the Epic Sepsis Model is early detection and improved outcomes in sepsis care. Ethical goals constrain how that purpose is achieved rather than overriding it. For primary stakeholders, functional goals significantly outnumber ethical constraints, reflecting the system's clinical purpose while ensuring ethical safeguards remain enforceable.

Functional Goals	Ethical Goals
Early detection	Prevent over-reliance (V1)
Threshold-based alerting	Preserve epistemic access (D1)
Reduced mortality	Enforce governance accountability (D2)
Workflow integration	Control false positive burden (U2)
Antibiotic timing improvement	Prevent clinical deskilling (V2)

Table 2.2: Functional vs. ethical goals of the ESM.

Ethical goals are more difficult to implement and validate but do not overshadow the system's clinical objectives.

Chapter 3: **Implementation**

Link to Repository: https://github.com/AshwinPrasanth/AI–Ethics
The repository contains a structured simulation and formal verification framework modelling the decision-structural properties of the Epic Sepsis Model. The objective is not architectural replication of the proprietary system, but formalisation and validation of functional and ethical properties. The implementation consists of two validation layers: a **behavioural simulation layer** and a **formal logical verification layer** (Z3 SMT). Together, they operationalise ethical evaluation as executable constraint verification.

3.1 Description of Approach

3.1.1 Model Abstraction

The Epic Sepsis Model is a proprietary, data-driven predictive system trained on approximately 500,000 patient encounters and incorporating ∼80 clinical variables. Its internal architecture, feature weighting, and training pipeline are not publicly disclosed.
This implementation does not attempt model reproduction. Instead, it preserves the operational invariants that define the system's decision structure: continuous numerical risk scoring; threshold-based alert triggering (Score ≥ 5); screening-oriented behaviour (high sensitivity design); non-zero false positive possibility; human-in-the-loop decision authority; and integration into clinician workflow. The system therefore represents a structural abstraction of a sub-symbolic CDSS, sufficient for ethical validation experiments.

3.1.2 Domain Representation

Each synthetic inpatient instance is represented as:

$$x = (T,\ HR,\ Lactate,\ WBC,\ SBP,\ PriorAntibiotics)$$

where T is temperature, HR heart rate, WBC white blood cell count, and SBP systolic blood pressure. These represent a clinically meaningful subset of ESM inputs. The risk score is computed as:

$$Score = \sum_i w_i x_i$$

with fixed weights w_i. The purpose is not predictive realism, but simulation of sub-symbolic weighted aggregation of heterogeneous physiological signals.

3.1.3 Functional Goal Encoding

Functional goals are encoded in `functional_validation.py`.

F1: Continuous Risk Stratification. The system produces a real-valued risk score for each patient. High-risk synthetic cases yield $Score \geq 5$; moderate-risk cases yield $Score < 5$.

F2: Deterministic Alert Trigger. $Alert = (Score \geq 5)$, aligning with the published ESM cut-point.

F3: Workflow Activation. If $Alert =$ True, the system simulates escalation to a clinician and a decision node requiring response, modelling the real-world operational role of ESM alerts.

3.1.4 Ethical Goal Formalisation (Simulation Layer)

Ethical goals are encoded as executable constraint checks in `ethical_validation.py`.

E1: False Positive Monitoring (Utilitarian Trade-Off). $Alert = \text{True} \wedge GroundTruth = \text{False}$. The False Positive Rate is computed as $FPR = FP/(FP + TN)$. If $FPR > 0$, alert fatigue risk is flagged and logged.

E2: Over-Reliance Detection (Automation Bias). $Alert = \text{True} \wedge ClinicianAction = Accept \wedge GroundTruth = \text{False}$. Captures blind compliance with AI output.

E3: Under-Reliance Detection. $Alert = \text{True} \wedge ClinicianAction = Ignore$ triggers $EthicalGoalSatisfied = \text{False}$.

E4: Responsibility Gap Detection. $Alert = \text{True} \wedge ExplanationUnavailable$. *Known limitation:* In the implementation, `generate_explanation()` always returns a feature contribution dictionary, so the gap is never triggered in simulation. The real ESM does not expose its internal reasoning; the Z3 layer addresses this structurally.

E5: Alert Fatigue Monitoring. If counter > 5, fatigue risk is flagged.

E6: Governance Constraint. $Alert \rightarrow Acknowledged$. Violation triggers a governance failure report.

E7: Data Quality Failure. Incomplete patient inputs trigger exceptions and logging. When `temperature: None` is passed, a `ValueError` is raised and written to `audit.log`.

E8: Age-Based Fairness Scenario. Two clinically equivalent patients differing only in age category are evaluated. If $EquivalentClinicalState \rightarrow (Alert_{elderly} = Alert_{young})$ is violated, subgroup disparity is flagged. Because the current scoring function does not incorporate age_group as a feature, the simulation always returns equivalent scores; the Z3 layer handles the formal violation case.

3.1.5 Formal Logical Verification (Z3 SMT Layer)

Selected ethical constraints are encoded as logical formulas and analysed using the Z3 SMT solver in `z3_ethics_verification.py`. Constraints are registered with `assert_and_track`, enabling UNSAT core extraction.

SAT Case: Duty-of-Care Consistency. $(Alert \wedge GroundTruth) \rightarrow ClinicianResponded$. Z3 returns SAT, confirming logical consistency.

UNSAT Case: Capacity Conflict. Two ethical mandates $(Treat_A \wedge Treat_B)$ conflict with the physical constraint $\neg(Treat_A \wedge Treat_B)$. Z3 returns UNSAT and extracts the minimal core: {`Ethical_Treat_A`, `Ethical_Treat_B`, `Capacity_Constraint`}.

UNSAT Case: Age-Based Fairness Violation. $EquivalentClinicalState \rightarrow (Alert_{elderly} = Alert_{young})$. Violation produces UNSAT; core isolates {`Elderly_Alert_True`, `Equivalent_Group_True`, `Fairness_Symmetry`, `Young_Alert_False`}.

UNSAT Case: Governance Violation. $Alert \rightarrow Acknowledged$. UNSAT core: {`Governance_Rule`, `Alert_Occurred`, `No_Acknowledgement`}.

Architectural significance. The SMT layer detects structural ethical contradictions, extracts minimal conflicting constraint sets, distinguishes behavioural violation from logical impossibility, and complements simulation-based evaluation.

3.1.6 Scope Delimitation

The implementation does not replicate proprietary ESM weighting, perform ROC optimisation, estimate AUC, model mortality causality, or use temporal logic. The system focuses on threshold-based decision structure and constraint consistency.

3.2 Points of Conflict

Conflict 1: Fairness vs. Clinical Equivalence

Original decision: The initial scoring model treated all patients identically using only physiological variables, with no subgroup differentiation.

Value conflict: Deontological and virtue ethics considerations raised the question of whether a system that does not explicitly test for subgroup equity could produce discriminatory outcomes for elderly patients with atypical sepsis physiology.

Design change: Two synthetic patient scenarios namely, `elderly_high_risk_patient()` and `young_high_risk_patient()` were introduced. A fairness check (E8) was added to the simulation layer and a formal fairness symmetry constraint was encoded in the Z3 layer.

Stakeholders who benefit: Patients (particularly elderly patients); regulators requiring evidence of non-discriminatory behaviour.

Residual limitation: The current scoring function does not use `age_group` as a feature, so the simulation always returns equivalent outcomes. The Z3 layer formally proves what would happen if unequal outcomes occurred.

Conflict 2: Transparency vs. Proprietary Opacity

Original decision: The abstraction included a `generate_explanation()` function returning feature contributions, treating interpretability as available.

Value conflict: This misrepresented the real ESM. Deontological concerns around accountability required the abstraction to reflect opacity rather than paper over it.

Design change: The E4 constraint was retained but documented as a known abstraction limitation. The Z3 layer was extended to formally model governance and accountability constraints that the simulation layer could not fully instantiate.

Stakeholders who benefit: Clinicians; patients whose safety depends on clinicians understanding the limits of AI-generated scores.

Conflict 3: Sensitivity Optimisation vs. Alert Fatigue

Original decision: The alert threshold was fixed at Score ≥ 5 without analysis of how threshold variation affects ethical outcomes.

Value conflict: Utilitarian analysis identified that optimising for sensitivity necessarily increases false positives and accumulates alert burden on clinicians, creating a conflict between population-level mortality benefit and individual workflow sustainability.

Design change: A threshold trade-off simulation was added, evaluating alert behaviour at thresholds of 4, 5, and 7. An alert fatigue counter (E5) and expanded audit logging were introduced.

Stakeholders who benefit: Clinicians; hospitals using threshold configuration as a governance lever; patients, indirectly.

3.3 Evidence

Functional Validation Output

Execution of `python functional_validation.py` produces:

```
@AshwinPrasanth → /workspaces/AI--Ethics (main) $ /usr/bin/python3 /workspaces/AI--Ethics/epic-ethical-validation/functional_validation.py

--- Functional Validation ---
High Risk Score: 5.750000000000001
Alert Triggered: True
Moderate Risk Score: 2.9800000000000013
Alert Triggered: False
```

Figure 3.1: Terminal output of `functional_validation.py`.

A high-risk profile scores 5.75, correctly triggering an alert. A moderate-risk profile scores 2.98, correctly suppressing it. This validates functional goals **F1** and **F2**.

Ethical Validation Output

Execution of `python ethical_validation.py` produces:

```
@AshwinPrasanth → /workspaces/AI--Ethics (main) $ /usr/bin/python3 /workspaces/AI--Ethics/epic-ethical-validation/ethical_validation.py

--- Over-reliance Scenario ---
Over-reliance detected: True

--- Basic False Positive Case ---
False Positive Rate: 0.0

--- Aggressive False Positive Case ---
Aggressive False Positive Score: 5.402000000000001
Alert Triggered: True
False Positive Rate: 1.0

--- Under-reliance Scenario ---
Under-reliance scenario:
Alert: True
Clinician responded: False
Ethical goal satisfied: False

--- Responsibility Gap Test ---
Alert: True
Explanation available: True
Responsibility gap detected: False

--- Threshold Trade-Off Simulation ---
Threshold: 4
Alert Triggered: True
False Positive Rate: 1.0
------------------------------
Threshold: 5
Alert Triggered: True
False Positive Rate: 1.0
------------------------------
Threshold: 7
Alert Triggered: False
False Positive Rate: 0.0

--- Reliance Calibration Test ---
Reliance type: Blind Compliance

--- Alert Fatigue Simulation ---
Alert fatigue triggered: True

--- Governance Check ---
Governance constraint satisfied: False

--- Data Quality Failure Scenario ---
Data quality failure detected: Missing required feature: temperature

--- Fairness Across Subgroups ---
Elderly Alert: True
Young Alert: True
No subgroup disparity detected.
```

Figure 3.2: Terminal output of `ethical_validation.py`.

Output	Ethical Goal	Result
Over-reliance detected: True	E2 — Automation Bias	Violation detected
Basic FPR: 0.0	E1 — FP Monitoring	Correct (below threshold)
Aggressive FPR: 1.0	E1 — FP Monitoring	Violation flagged
Ethical goal satisfied: False	E3 — Under-reliance	Violation detected
Responsibility gap: False	E4 — Responsibility Gap	Abstraction limit
Alert fatigue: True	E5 — Alert Fatigue	Violation flagged
Governance satisfied: False	E6 — Governance	Violation detected
Data quality failure	E7 — Data Quality	Exception caught
No subgroup disparity	E8 — Fairness	Z3 handles formal case

Table 3.1: Ethical validation results mapped to ethical goals.

Z3 Formal Verification Output

Execution of `python z3_ethics_verification.py` produces:

```
@AshwinPrasanth → /workspaces/AI--Ethics/epic-ethical-validation (main) $ python3 z3_ethics_verification.py

--- SAT: Duty of Care Scenario ---
Solver result: sat
Model: [Alert = True,
 ClinicianResponded = True,
 Duty_of_Care = True,
 GroundTruth = True]

--- UNSAT: Capacity Conflict ---
Solver result: unsat
Unsat Core: [Ethical_Treat_A, Ethical_Treat_B, Capacity_Constraint]

--- UNSAT: Fairness Violation ---
Solver result: unsat
Unsat Core: [Elderly_Alert_True,
 Equivalent_Group_True,
 Fairness_Symmetry,
 Young_Alert_False]

--- UNSAT: Governance Violation ---
Solver result: unsat
Unsat Core: [Governance_Rule, No_Acknowledgement, Alert_Occurred]
```

Figure 3.3: Terminal output of `z3_ethics_verification.py`.

The SAT result confirms the ethical rule set is consistent under compliant behaviour. The three UNSAT results prove that fairness violations, triage deadlock, and governance failures are logically impossible to reconcile with their respective constraints. UNSAT core extraction identifies the minimal responsible constraint sets.

Audit Log

```
audit.log
2026-02-11 20:08:28,079 - WARNING - Over-reliance scenario detected.
2026-02-11 20:08:28,079 - INFO - False Positive Test - Score: 5.402000000000001, Alert: True, FPR: 1.0
2026-02-11 20:08:28,080 - WARNING - Under-reliance detected - Alert: True, Clinician responded: False
2026-03-09 10:52:32,101 - WARNING - Over-reliance scenario detected.
2026-03-09 10:52:32,111 - INFO - False Positive Test - Score: 5.402000000000001, Alert: True, FPR: 1.0
2026-03-09 10:52:32,111 - WARNING - Under-reliance detected - Alert: True, Clinician responded: False
2026-03-09 10:52:32,111 - ERROR - Data quality failure detected: Missing required feature: temperature
2026-03-09 18:59:45,694 - WARNING - Over-reliance scenario detected.
2026-03-09 18:59:45,694 - INFO - False Positive Test - Score: 5.402000000000001, Alert: True, FPR: 1.0
2026-03-09 18:59:45,695 - WARNING - Under-reliance detected - Alert: True, Clinician responded: False
2026-03-09 18:59:45,695 - ERROR - Data quality failure detected: Missing required feature: temperature
```

Figure 3.4: Contents of `audit.log`.

The audit log provides a persistent, timestamped accountability trace across multiple runs. Repeated entries confirm that synthetic scenarios are deterministic by design. The log records over-reliance warnings, false positive events, under-reliance violations, and data quality failures, modelling the audit trail a real deployed CDSS would be expected to maintain for governance purposes.

Chapter 4: **Claude Says**

LLM Details

Model used: Claude Sonnet (claude.ai, free tier)
Date of debate: March 2026

All responses were generated using the free tier interface at claude.ai. No system prompt modifications were used. Each argument was presented in a new conversation to prevent contextual carry-over between responses.

The objective of this exercise was not simply to obtain answers but to examine the logical structure of the reasoning produced by the model. Where responses contained logical weaknesses or incomplete reasoning these were identified explicitly and analysed.

Overview

This chapter records a structured debate with Claude across the four ethical conflict points identified in Chapter 3. Each argument was prompted separately and Claude was asked to provide at least two positions together with reasoning.

The responses are reproduced and then examined critically. Logical fallacies are identified where they occur and the stronger position is selected based on evidence from the literature and the ethical framework developed earlier in this report.

Argument 1: Sensitivity Optimisation vs. Alert Fatigue

Should the Threshold Be Raised?

Prompt: In a deployed AI sepsis detection system, the alert threshold is set at Score ≥ 5, optimising for high sensitivity (86%) but producing a positive predictive value of only 33.8%. This means roughly two thirds of alerts are false positives. From an ethical standpoint, should the threshold be raised to reduce alert fatigue, even if some sepsis cases are missed? Give me at least two distinct actions or non-actions you would recommend, and explain the reasoning behind each. Please keep your response to 300–400 words total.

Figure 4.1: Claude's Prompt to Argument 1

Claude's Response — Ethical Analysis: Sepsis Alert Threshold

Recommendation 1: Do not raise the threshold unilaterally — instead, stratify alerts by context. Rather than a blunt threshold increase, the system should implement tiered alerting based on clinical setting. An ICU team managing critically ill patients can act on lower-confidence alerts; a general ward with stretched staffing cannot. Deploying a higher threshold (say, ≥ 7) only in lower-acuity settings preserves sensitivity where it matters most while reducing noise where alert fatigue is most damaging. The ethical justification is proportionality.

Recommendation 2: Measure and act on alert fatigue empirically before changing the threshold at all. Before adjusting the threshold, the deploying organisation should audit clinician override rates, response times, and documented sepsis cases acted upon versus ignored. If clinicians are already dismissing 70% of alerts reflexively, the current threshold may already be functionally equivalent to a higher one — just with hidden harm. The ethical principle here is epistemic humility.

Underlying principle: Optimising a threshold is not purely a statistical exercise — it is a values-laden decision about whose risk is acceptable and who bears the burden of error. That decision should involve clinicians, patients, and ethicists, not only the AI development team.

Figure 4.2: Claude's response to Argument 1

Ethical Principle Evaluated.

School: Utilitarianism (Mill, principle of harm minimisation)
Constraint: The burden of false positive alerts must remain within operational tolerance so that clinicians remain responsive to genuine warnings. Specifically, if the clinician override rate exceeds 70 percent across consecutive alert sessions, the system is no longer producing net benefit (Goal U2, Chapter 2).
Violation condition: If false positive alerts become sufficiently frequent that clinicians ignore them reflexively, the system undermines patient safety rather than supporting it.

Analysis. Recommendation 2 is the stronger position. Decisions about threshold adjustment should be grounded in empirical evidence about clinician response behaviour. Shwedeh and Alzoubi [3] demonstrate that over-reliance and alert fatigue emerge specifically in high-pressure deployment environments where governance is insufficient, reinforcing that threshold decisions must be informed by local usage data rather than modelling assumptions alone.
Recommendation 1 contains a logical flaw. The proposal assumes that ward type reliably reflects the consequences of false positives and false negatives. A patient with a risk score of six on a general ward has the same physiological condition as a patient with the same score in an intensive care unit. Under this proposal only one of those patients would generate an alert. General wards also typically have less continuous monitoring than intensive care units, which may make missed deterioration more dangerous rather than less. The logic of proportionality runs in the opposite direction to what Claude claims.

Fallacy identified: Hasty Generalisation. Ward location is treated as a reliable proxy for patient risk when the relationship is not justified. Because the proportionality claim depends on that assumption the argument collapses. Recommendation 2 is therefore the stronger ethical position.

Argument 2: Transparency vs. Proprietary Opacity

Is the Responsibility Gap Defensible?

Prompt: The Epic Sepsis Model generates a risk score but does not expose its internal reasoning to clinicians. Clinicians remain legally and morally accountable for decisions made in response to AI alerts, yet cannot fully explain the basis of those alerts. Is this ethically acceptable? Give me at least two distinct positions –- one defending the current opaque design and one arguing for mandatory explainability –- with the reasoning behind each. Please keep your response to 300–400 words total.

Figure 4.3: Claude's Prompt to Argument 2

Claude's Response — Explainability in the Epic Sepsis Model: Two Positions

Position 1: Opacity is defensible — clinicians are not obligated to understand the tool's internals. A clinician using an opaque sepsis alert is in a structurally similar position to one interpreting a lab result: they do not need to understand the chemistry of a troponin assay to act appropriately on an elevated reading. If clinicians know the score has 86% sensitivity and a 33.8% PPV, they have actionable calibration information. Demanding full mechanistic transparency may be counterproductive — complex feature interactions can produce explanations that are technically accurate but cognitively misleading.

Position 2: Opacity is ethically unacceptable when accountability remains with the clinician. Clinicians bear full legal and moral responsibility for outcomes, yet are denied access to the reasoning that generated the alert. This is not analogous to a lab test — a troponin value is a direct measurement of a biological quantity, whereas an AI score is an inference built on weighted feature combinations that may encode biases or training artefacts invisible to the end user. Meaningful accountability requires meaningful access.

Unresolved tension: Both positions share a concern for patient safety but disagree on whether opacity or false explainability poses the greater risk. That disagreement demands empirical study of how clinicians actually use explanations when provided.

Figure 4.4: Claude's response to Argument 2

Ethical Principle Evaluated.

School: Deontological ethics (Kant, respect for rational agency)
Constraint: A clinician who is held legally and morally responsible for a clinical decision must have meaningful epistemic access to the basis of the information that prompted that decision. This is formalised as Goal D1 in Chapter 2. The system satisfies this principle only if, for every alert fired, the clinician can access at minimum the top contributing feature variables and their directional influence on the score.
Violation condition: Any alert fired without accompanying feature-level explanation constitutes a violation. In the current ESM deployment this condition is violated for 100 percent of alerts.

Analysis. Position 2 identifies the central issue correctly. Clinicians remain accountable for outcomes yet are denied access to the reasoning behind the alerts they must interpret. Amann et al. [4] demonstrate that lack of explainability causes systems nominally classified as decision support to drift toward decision determining behaviour, undermining genuine human oversight.

Čartolovni et al. [6] identify the resulting accountability gap as the dominant ethical concern in the AI-based medical decision support literature.
Position 1 depends on an analogy with laboratory tests. This comparison is flawed. A troponin value is a direct biological measurement with known, traceable error properties. An AI-generated risk score is a learned inference derived from correlations across many variables and may encode biases or data artefacts invisible to the clinician [2]. Xu et al. [5] establish that post-hoc explanations and population-level metrics are insufficient to guarantee accountability in sub-symbolic clinical decision support systems. When the ESM is systematically wrong for a particular patient group, that error is invisible at the point of care.

Fallacy identified: False Analogy. The analogy transfers the epistemic legitimacy of laboratory measurements onto a fundamentally different class of predictive system. Because the analogy is the foundation of Position 1 the defence of opacity collapses with it. Position 2 is therefore the stronger position.

Argument 3: Fairness Across Age Subgroups

Should Testing Be Mandatory Pre-Deployment?

> Prompt: A sepsis risk model is trained on a large patient population but does not explicitly test for performance differences across age groups. Elderly patients often present with atypical sepsis physiology and may be systematically underscored. Should fairness testing across age subgroups be a mandatory requirement before deploying such a system? Give me at least two distinct actions or non-actions, with reasoning for each. Please keep your response to 300–400 words total.

Figure 4.5: Claude's Prompt to Argument 3

> **Claude's Response — Fairness Testing Across Age Subgroups: Two Positions**
>
> **Action 1: Mandate subgroup fairness testing as a pre-deployment requirement.** Deploying a model without testing performance across age groups is not a neutral omission — it is a decision to accept unknown differential harm. Elderly patients present particular physiological challenges: blunted fever responses, atypical inflammatory markers, and baseline comorbidities that may systematically suppress risk scores. The ethical principle of justice requires that foreseeable disparate impacts be identified before deployment, not discovered retrospectively through adverse outcomes.
>
> **Non-action 2: Do not delay deployment pending perfect subgroup parity — but require post-deployment monitoring.** Mandatory pre-deployment fairness testing risks becoming a barrier that keeps demonstrably useful tools out of clinical practice. A model with imperfect subgroup performance may still reduce net sepsis mortality compared to no decision support at all. The more proportionate requirement is conditional deployment alongside a mandatory post-deployment surveillance protocol stratified by age.
>
> **Shared commitment:** Both recommendations reject deploying without any fairness consideration. What is ethically indefensible is treating subgroup performance as an afterthought
> — something to investigate only after a pattern of missed diagnoses forces the question.

Figure 4.6: Claude's response to Argument 3

Ethical Principle Evaluated.

School: Deontological ethics (principle of equal treatment) and Utilitarian justice
Constraint: Patients with equivalent clinical states must receive equivalent diagnostic evaluation regardless of demographic subgroup. This is formalised in Chapter 2 and operationalised in the Z3 fairness constraint as: $EquivalentClinicalState \rightarrow (Alert_{elderly} = Alert_{young})$. If this condition is violated it breaches the principle of equal treatment under Goal D1.
Violation condition: Any systematic difference in alert rates between clinically equivalent patients in different age subgroups constitutes a violation detectable before deployment.

Analysis. Action 1 reflects the justice principle directly. Deploying without subgroup testing accepts unknown differential harm. Elderly patients are not an edge case in sepsis care and the physiological reasons their scores may be suppressed are well documented clinically [2].
Action 2 relies on an incorrect comparison class. The alternative to a fairness-tested model is not the absence of clinical decision support. Most hospitals already operate structured sepsis screening protocols and early warning scores. Vasey et al. [1] explicitly caution that ethical evaluation must compare against the existing standard of care rather than against no intervention.
A second problem is that post-deployment monitoring collects evidence from patients already being treated under the untested system. Čartolovni et al. [6] identify this as the accountability gap in AI-based medical decision support, noting that deferring fairness assessment to post-deployment monitoring makes institutional responsibility permanently ambiguous during the observation period.

Fallacy identified: False Cause. Action 2 implies that fairness violations can only be discovered after deployment. The formal Z3 verification layer in this project demonstrates that subgroup parity constraints can be logically checked before any patient is involved. The UNSAT core output directly refutes the causal claim that post-deployment monitoring is the only feasible approach. Action 1 is therefore the stronger ethical position.

Argument 4: Automation Bias

Design Problem, Training Problem, or Inherent Risk?

Prompt: Studies show that when clinicians are repeatedly exposed to AI alerts, they tend to follow them without independent assessment –- a phenomenon called automation bias. In a high-stakes clinical setting like sepsis detection, is automation bias primarily a training problem, a system design problem, or an inherent risk of deploying AI in clinical workflows at all? Give me at least two distinct recommended actions or non-actions, with the reasoning behind each. Please keep your response to 300–400 words total.

Figure 4.7: Claude's Prompt to Argument 4

Claude's Response — Automation Bias in Sepsis AI: Diagnosis and Response

Position 1: Automation bias is primarily a system design problem — and design can mitigate it. A binary alarm invites passive compliance; a display that shows contributing factors and prompts documentation of independent clinical reasoning invites engagement. Interface friction — requiring a clinician to actively confirm or override with a brief rationale — reduces reflexive compliance without meaningfully slowing response times. If automation bias is reliably produced by particular interface patterns, deploying those patterns unreflectively is an institutional choice, not an inevitability.

Action 2: Implement structured training, but treat it as necessary rather than sufficient. Training clinicians to recognise automation bias and practice deliberate override decisions is ethically mandatory. However, cognitive shortcuts under time pressure and fatigue are predictable features of human performance in high-load environments, not failures of knowledge. Training addresses the skill gap; it does not address the conditions that make bias adaptive in the first place.

Unresolved question: Automation bias may be partially inherent to any system where AI output is faster and less effortful than independent assessment. The honest institutional response is to explicitly accept residual bias as a known risk and maintain human staffing levels sufficient to make genuine clinical reasoning possible.

Figure 4.8: Claude's response to Argument 4

Ethical Principle Evaluated.

School: Virtue ethics (Aristotle, phronesis and professional competence)
Constraint: Clinical decision support must assist rather than replace independent clinical reasoning. This is formalised as Goals V1 and V2 in Chapter 2. Automation bias is operationally present when clinician compliance with alerts exceeds 90 percent without documented independent assessment. Deskilling risk is flagged when independent override decisions fall below a minimum threshold per clinician per month.
Violation condition: Compliance rate greater than 90 percent without independent documentation, or a sustained reduction in clinician independent override frequency over time.

Analysis. The distinction between interface design and training is reasonable. Panigutti et al. [7] provide empirical evidence that explanations and interface changes do influence reliance on AI advice, though not always in ways that improve decision accuracy. Design and training therefore address different failure modes and both are necessary.
However neither position engages with the deeper virtue ethics concern. Automation bias is treated throughout as a cognitive shortcut problem. Čartolovni et al. [6] identify deskilling and loss of epistemic authority as physician-specific risks that are distinct from momentary automation bias. A clinician who has repeatedly followed ESM alerts without practising independent diagnostic assessment may over time lose the calibrated clinical judgment that makes meaningful override possible. This is Goal V2 in Chapter 2 and neither of Claude's positions addresses it.
The closing observation that automation bias may be inherent is the most ethically significant point in the response. Claude identifies the problem and then offers no conclusion from it.

Fallacy identified: Argument from Ignorance. The absence of a definitive solution to inherent automation bias is implicitly treated as justification for not requiring one. Both positions therefore remain incomplete because neither addresses the long-term professional deskilling risk that virtue ethics identifies as the deeper problem.

Summary of Identified Errors

Argument	Claude Position	Fallacy	Supported Position
Threshold decision	Ward based threshold stratification	Hasty Generalisation	Empirical audit before any threshold change
Model opacity	Analogy with laboratory tests	False Analogy	Explainability required when responsibility remains with the clinician
Age fairness	Conditional deployment with monitoring	False Cause	Mandatory pre-deployment subgroup fairness testing
Automation bias	Design and training sufficient	Argument from Ignorance	Additional safeguards required including explicit deskilling monitoring

Table 4.1: Summary of logical errors identified in Claude's responses.

Chapters 5 and 6 each take one of these fallacies and argue against Claude's position using evidence from the repository.

Chapter 5: Teammate 1 Says

Selected Argument

This chapter argues against Claude's Position 1 in Argument 2, which claimed that opacity in the Epic Sepsis Model is ethically defensible because clinicians interpreting an AI risk score are in the same position as clinicians interpreting a laboratory result. This argument rests on a false analogy fallacy. Because the fallacy undermines the entire defence of opacity, Position 2 that opacity is ethically unacceptable when accountability remains with the clinician is the correct position.

Ethical Principle

School: Deontological ethics (accountability and responsibility)
Principle: A clinician who is held legally and morally responsible for a clinical decision must have meaningful epistemic access to the basis of the information that informed that decision.
Operational constraint: If an AI system generates a clinical alert, the clinician must be able to access the reasoning or contributing factors that produced the alert.
Violation condition: If the system produces alerts without any explanation or accessible reasoning, the clinician's responsibility cannot be justified. Under these conditions the system creates a responsibility gap between formal authority and practical epistemic control.

The Fallacy: False Analogy

Claude's Position 1 draws a structural equivalence between interpreting an ESM alert and interpreting a troponin result. The argument is that in both cases the clinician acts on a tool output without understanding its internal mechanics, and that this is considered acceptable practice. The analogy fails because the two situations have categorically different error structures.

A troponin result is a direct measurement of a biological quantity with documented, investigable failure modes: reagent degradation, sample haemolysis, assay interference from skeletal muscle troponin. When a troponin is anomalous or inconsistent with clinical presentation, the clinician and the laboratory can reason about why. The source of error is traceable and, in principle, correctable.

The ESM score is a learned inference from a proprietary model trained on approximately 500,000 historical encounters, combining roughly 80 variables through undisclosed statistical relationships [2]. Those relationships may encode documentation biases, historical care inequities, or population-specific confounders. When the ESM is systematically wrong for a particular patient group, that error is invisible at the point of care. Xu et al. [5] establish that this opacity is not a minor limitation but a fundamental ethical problem: post-hoc explanations and population-level metrics are insufficient to guarantee accountability in sub-symbolic CDSS. There is no equivalent of checking the reagent batch. The clinician receives a number with no accessible basis for disagreement.

Equating these two situations is a false analogy. One error is traceable. The other is structurally inaccessible.

What the Fallacy Conceals

The practical consequence of accepting Claude's false analogy is that the responsibility gap is rendered invisible. Amann et al. [4] demonstrate that lack of explainability causes systems nominally classified as decision support to drift toward decision determining the clinician retains formal authority while practical epistemic control has shifted to the model. Claude's analogy imports the legitimacy of an established diagnostic measurement onto a system with fundamentally different accountability implications, concealing this drift.

Consider the scenario the ESM is most likely to produce: a clinician receives an alert for an elderly patient with atypical presentation, assesses the patient independently, and decides the score does not reflect the clinical picture. They document their reasoning and do not escalate. The patient later deteriorates. The clinician's defence requires articulating not only what they observed but why they disagreed with the score. Under an opaque system, the second part of that defence has no grounding. This is not analogous to acting on a troponin without knowing the biochemistry. It is acting on an inference without knowing what was inferred or why precisely the condition Čartolovni et al. [6] identify as the dominant accountability gap in AI-based medical decision support.

The Structural Nature of the Gap

The responsibility gap is therefore not an exceptional scenario but a structural feature of the deployment architecture, not an edge case. The E4 constraint formalises this directly:

```
def responsibility_gap(alert, explanation_available):
    if alert and not explanation_available:
        return True
    return False
```

In current published descriptions of the ESM deployment, clinicians are not provided with feature-level explanations for alerts, meaning `explanation_available` is effectively `False` in practice. The function therefore returns `True` whenever an alert is triggered. Claude argues that population-level metrics such as 86% sensitivity and 33.8% PPV provide sufficient calibration. They do not. Population statistics describe aggregate model behaviour, they say nothing about the basis for any individual alert, which is what a clinician needs to exercise the independent judgement that accountability requires.

An Internal Contradiction

Claude's Position 1 also contradicts what Claude acknowledged in Argument 4. The defence of opacity depends on clinicians consciously using population-level metrics to calibrate their response to each alert. But in Argument 4, Claude accepted that clinicians under time pressure and fatigue default to following alerts without independent assessment. Panigutti et al. [7] provide empirical confirmation: explanations systematically increase human reliance on AI advice regardless of whether reliance improves accuracy, meaning the calibrated reflective clinician that Position 1 assumes is not the clinician that real deployment produces. These two pictures of clinical behaviour are incompatible. Claude cannot use the first to justify opacity while accepting the second as the operational reality.

Conclusion

The lab test analogy is a false analogy fallacy. It treats traceability of error as irrelevant to accountability when it is central to it. Because the fallacy is the load-bearing element of Position 1, the entire defence of opacity collapses with it. Position 2 is correct: opacity is ethically unacceptable when clinicians bear full accountability for decisions whose basis they cannot access. The responsibility gap is not a theoretical concern, it is a structural feature of every alert the system fires.

Chapter 6: Teammate 2 Says

Selected Argument

This chapter argues against Claude's Non-action 2 in Argument 3, which proposed replacing mandatory pre-deployment fairness testing with conditional deployment accompanied by post-deployment monitoring. This argument contains a false cause fallacy and it attributes the impossibility of pre-deployment fairness testing to an epistemic limitation that does not exist. Because the fallacy is the basis for the entire conditional deployment position, Action 1 mandating subgroup fairness testing as a pre-deployment requirement is the correct position.

Ethical Principle

School: Deontological ethics (principle of equal treatment) with utilitarian justice considerations.
Principle: Patients with equivalent clinical states must receive equivalent diagnostic evaluation regardless of demographic subgroup.
Operational constraint: If two patients exhibit equivalent clinical indicators for sepsis, the system must produce equivalent alert decisions regardless of age group.
Violation condition: If two clinically equivalent patients receive different alert outcomes solely due to subgroup membership, the fairness constraint is violated.

The Fallacy: False Cause

Claude's Non-action 2 implies that the reason for deferring fairness testing to the post-deployment phase is that the violation cannot be known in advance. The argument runs: a model with imperfect subgroup performance may still reduce net sepsis mortality compared to no decision support, so conditional deployment with post-deployment surveillance is a proportionate response to an uncertainty that cannot be resolved before going live.
This is a false cause fallacy. The claimed cause of deferral epistemic impossibility is not the actual cause. Fairness violations are formally detectable before deployment. The Z3 verification layer in this project demonstrates this directly:

```
def unsat_fairness_violation():
    s = z3.Solver()
    Equivalent_Group = z3.Bool("Equivalent_Group")
    elderly_alert = z3.Bool("elderly_alert")
    young_alert = z3.Bool("young_alert")

    s.assert_and_track(
        z3.Implies(Equivalent_Group, elderly_alert == young_alert),
        "Fairness_Symmetry"
    )
    s.assert_and_track(Equivalent_Group, "Equivalent_Group_True")
    s.assert_and_track(elderly_alert, "Elderly_Alert_True")
    s.assert_and_track(z3.Not(young_alert), "Young_Alert_False")

    result = s.check()
    if result == z3.unsat:
        print("Unsat Core:", s.unsat_core())
```

This returns UNSAT with core `[Elderly_Alert_True, Equivalent_Group_True, Fairness_Symmetry, Young_Alert_False]`. The violation scenario is formally proven to be logically inconsistent with the fairness constraint before any patient is involved. Xu et al. [5] further identify the absence of formal verification as a specific research gap in AI-based CDSS ethics, noting that ethical compliance is seldom enforced through logic-based constraints which is precisely what this implementation provides. The epistemic precondition for Claude's conditional deployment argument is false.

The Wrong Counterfactual

Claude's argument also rests on a second error. The claim that a model with imperfect subgroup performance may still reduce net sepsis mortality compared to no decision support at all uses the wrong comparison class. The alternative to a fairness-tested model is not an empty clinical environment. Most hospital systems deploying the ESM already operate structured sepsis screening protocols, early warning scores, and nursing assessment frameworks [3]. Vasey et al. [1] (DECIDE-AI) explicitly caution that high retrospective accuracy does not guarantee clinical benefit or safety in real-world deployment, and that ethical evaluation must account for the existing standard of care rather than a zero baseline. The net benefit calculation Claude implies only holds if the counterfactual is nothing, which it is not.

Post-Deployment Monitoring Is Not a Safeguard

Having removed the epistemic justification for deferral and corrected the counterfactual, what remains of Non-action 2 is a governance process that collects evidence of harm from patients who are already being managed under the untested system. Čartolovni et al. [6] identify this accountability gap directly: responsibility for harm in AI-based decision support is often unclear across clinicians, developers, and institutions, and frameworks that defer fairness assessment to post-deployment monitoring effectively make that ambiguity permanent during the observation period. The patients harmed during monitoring are not data points in a quality improvement exercise. They are people who did not receive a timely alert because the institution chose not to test for the disparity before deployment.

The justice principle identified under Goal D1 in Chapter 2 requires that foreseeable disparate impacts be identified before they occur, not after. Elderly patients present particular physiological challenges like blunted fever responses, atypical inflammatory markers, baseline comorbidities that may systematically suppress ESM scores [2]. This is not an unpredictable edge case. It is a foreseeable risk that the pre-deployment formal verification layer in this project is specifically designed to detect.

Conclusion

Claude's Non-action 2 rests on a false cause fallacy: the claim that fairness violations cannot be known before deployment. The Z3 verification layer in this project formally refutes that claim. With the fallacy removed, the conditional deployment position has no epistemic justification, uses the wrong counterfactual, and treats foreseeable harm to elderly patients as an administrative cost. Action 1 is correct, subgroup fairness testing must be a pre-deployment requirement, not a post-deployment observation.

Chapter 7: **Conclusions**

This project examined the Epic Sepsis Model through three ethical frameworks: utilitarianism, deontological ethics, and virtue ethics. Each framework illuminated a different dimension of the system's ethical profile. Utilitarian analysis surfaced the tension between sensitivity optimisation and false positive burden. Virtue ethics raised the longer term concern about automation bias and clinical deskilling. Deontological ethics identified the accountability and governance failures that sit at the core of the system's design.

Of the three, deontological ethics proved the most tractable for both expressing the problems and implementing solutions. The reason is structural. Deontological reasoning operates through obligations and rules: if an alert fires, it must be acknowledged; if a clinician is accountable, they must have epistemic access; if two patients are clinically equivalent, they must receive equivalent treatment. These translate directly into executable constraints. Every ethical goal in the implementation, from the responsibility gap check to the governance constraint to the fairness symmetry rule, is a formalised duty of this kind. The Z3 verification layer makes this particularly clear: encoding an ethical obligation as a logical formula and checking whether it is satisfiable is deontological reasoning in a formal system. Utilitarianism was useful for framing population level trade-offs, particularly around threshold configuration and false positive rates, but harder to implement rigorously. Meaningful utilitarian validation would require outcome data across real patient populations, which a simulation abstraction cannot provide. The threshold trade-off simulation approximates this but cannot quantify net mortality benefit, which limits how far the utilitarian analysis can be taken within this scope. Virtue ethics raised the most important long term concern, that repeated AI assisted workflows may erode the professional judgment that makes meaningful human oversight possible, but was the hardest to encode. Deskilling is a dynamic process that unfolds over years of clinical practice. It does not reduce naturally to a binary constraint or a satisfiability check, which meant it remained at the level of analysis rather than implementation. The broader lesson from this project is that ethical goals in AI systems are most effectively addressed when they are treated as first class design constraints rather than post-hoc commentary. The cases where the implementation fell short, particularly around the responsibility gap and the fairness simulation, were precisely the cases where the real system's opacity made formal constraint verification difficult. That difficulty is itself an ethical signal.

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